The Tightness Paradox: Hiring Friction, Measured Tightness, and Real Wages Across U.S. States, 2015–2023

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Abstract

We ask whether the gap between posted job openings and actual hires — hiring friction — drives real wages, using a newly assembled state-by-month panel of the 25 largest U.S. states from April 2015 to December 2023 (N = 2,625). A friction proxy, defined as unfilled openings per worker and lagged one quarter, predicts real wages within states over time: in a two-way fixed-effects specification with standard errors clustered by state, a one-percentage-point increase in the friction proxy is associated with roughly 6 percent higher real wages ($p < 0.05$). Yet the same relationship reverses sign in pooled OLS, where high-friction states pay less — the tightness paradox. Two follow-up tests return nulls: hiring friction does not explain the conventional tightness ratio (openings per unemployed worker), which instead loads on service-sector concentration; and the friction–wage relationship shows no detectable break around the pandemic. The results suggest that headline "labor shortage" tightness and hiring friction are distinct phenomena, and that the former is a noisier signal than commonly assumed.

1. Introduction

In the aftermath of the pandemic, the U.S. labor market looked historically tight: job openings exceeded the number of unemployed workers by a wide margin, and employers reported acute difficulty hiring. Yet wage growth was uneven, quit rates behaved strangely, and a growing share of postings appeared to be "ghost jobs" — openings posted with little intent to hire. This raises a measurement question with real stakes: does the standard tightness indicator, the ratio of openings to unemployed workers (V/U), capture genuine labor demand, or does it partly reflect postings that will never convert into hires?

This paper studies the gap between openings and hires directly. We construct a hiring-friction proxy — unfilled openings per worker — and ask three questions. First, does friction predict real wages? Second, does friction explain measured tightness, i.e., is the "shortage" a people shortage or a matching problem? Third, did the pandemic permanently change the friction–wage relationship?

The headline finding is a paradox. Across states, higher average friction is associated with lower average wages. But within a state over time — the comparison that strips out permanent state differences — rising unfilled openings predict rising real wages. The sign reversal between pooled and fixed-effects estimates is the paper's namesake: the tightness paradox. Two mechanism tests then come back null, which we report as faithfully as the positive result.

2. Data

We assemble a balanced state-by-month panel covering the 25 largest U.S. states from April 2015 through December 2023: 25 states × 105 months = 2,625 observations with no missing values. Openings and hires come from the BLS Job Openings and Labor Turnover Survey (JOLTS); labor force and unemployment from BLS Local Area Unemployment Statistics (LAUS); hourly earnings from BLS Current Employment Statistics (CES). Nominal wages are deflated with the national CPI for All Urban Consumers (FRED series CPIAUCSL), expressed in January-2015 dollars. Educational attainment — the share of adults 25+ with a bachelor's degree or higher — is a static state-level measure from the 2019 American Community Survey. The effective state minimum wage (the higher of the federal floor and the state legislated minimum) enters as a control for legislated wage compression.

Our key constructs are:

Friction proxy. (Openings − Hires) × 1,000 / Labor Force: the share of the labor force represented by postings that did not convert into hires in a given month. Positive values mean openings went unfilled. Sample mean: 1.16% (sd 1.22%).

Friction_Lag3. The friction proxy lagged three months within each state, capturing wage stickiness — firms adjust posted salaries with roughly a one-quarter delay.

Friction × Education. Interaction of the lagged friction proxy with log bachelor's attainment. The "Experience Wall" hypothesis holds that friction's wage effect

should be strongest in educated states, where unfilled postings reflect skill mismatch rather than churn.

Labor tightness. Openings / Unemployed workers, the standard Beveridge-curve tightness ratio; values above one indicate more postings than job seekers.

Real wage. CPI-deflated average hourly earnings; the log real wage (sample mean 3.24, ≈ $25.50/hour in January-2015 dollars) is the primary outcome.

Separately, we compile a national descriptive series estimating the share of JOLTS openings attributable to "ghost" postings — openings with no corresponding hiring intent — from September 2017 through early 2026. The estimated ghost share runs 2–5% before 2021, spikes to 10–19% during 2021–2023, then eases to roughly 7–9% by 2025–2026. We treat this series as descriptive context only; its construction methodology is not documented in this project, so it does not enter the regressions.

3. Empirical strategy

The analysis proceeds in four stages.

Stage 1 — the wage signal. We regress log real wages on the lagged friction proxy, the friction–education interaction, the unemployment rate, log minimum wage, and the leisure-and-hospitality employment share (a control for service-heavy economies). We estimate three specifications — pooled OLS, state fixed effects, and two-way (state and month) fixed effects — with standard errors clustered by state throughout. The TWFE specification is preferred: it isolates within-state, within-month variation.

Stage 2 — the structural mechanism. We regress log tightness (V/U) on the friction variables, asking whether unfilled postings explain the "feeling" of shortage. If the labor shortage is a matching problem rather than a headcount problem, friction should load strongly here.

Stage 3 — the broken signal. We split the sample at January 2021 and re-estimate the TWFE wage model on each subsample, testing whether the pandemic structurally changed the friction–wage relationship.

Stage 4 — the inefficiency tax. A back-of-the-envelope magnitude exercise translating the estimated friction coefficient into an annual dollar wage premium per 1,000 employees. We flag upfront that this exercise inherits the statistical uncertainty of Stage 3 and should be read as illustrative.

4. Results

4.1 Friction predicts wages — but the sign depends on the comparison

Table 1 reports the Stage 1 estimates. In the preferred TWFE specification, the lagged friction proxy enters positively and significantly: a one-percentage-point increase in the share of unfilled openings predicts roughly 6 percent higher real wages within a state (coefficient 6.17, $p < 0.05$). The unemployment rate enters positively but small (0.004, $p < 0.05$); the minimum wage and service-share controls are insignificant in the TWFE specification.

The striking feature of Table 1 is the sign reversal across specifications. In pooled OLS, friction enters at −38.63 ($p < 0.01$): states with persistently more unfilled openings pay lower wages on average. Adding state fixed effects flips the coefficient to +11.05 ($p < 0.01$), and adding time effects leaves it at +6.17 ($p < 0.05$). The cross-sectional correlation is confounded by permanent state differences — low-wage, high-churn states structurally post more openings per hire — while the within-state estimates capture what happens when a given state's hiring friction rises: wages follow, with a quarter's delay. That reversal is the tightness paradox.

The "Experience Wall" interaction — the hypothesis that friction's wage effect concentrates in educated states — is not supported in the preferred specification. It enters negatively and is only marginally significant in TWFE (−1.69, $p < 0.10$), and its sign likewise reverses from pooled OLS (+10.85, $p < 0.01$). Whatever skill-mismatch story the interaction was meant to capture does not survive state and time fixed effects.

*Table 1 — Stage 1: log real hourly wage on hiring friction. Standard errors clustered by state. *** p<0.01, ** p<0.05, * p<0.10.*

Variable	Pooled OLS	State FE	TWFE
Friction_Lag3	-38.6265***	11.0502***	6.1657**
Friction × Education	10.8514***	-3.0410***	-1.6916*
Unemployment rate	0.0055	0.0034**	0.0040**
ln(Min. wage)	0.2741***	0.0564***	-0.0112
Leisure & hosp. share	226.1854	-368.1254**	23.8548
R^2	0.5745	0.3094	0.0755
N	2,625	2,625	2,625
State FE	No	Yes	Yes
Time FE	No	No	Yes

Figure 1 plots the TWFE marginal effects with 95% confidence intervals; Figure 2 maps states by average friction and average real wage. The quadrant view makes the cross-sectional paradox visible: knowledge hubs (Massachusetts, Washington, Connecticut) combine high friction with high wages, while entry-level hubs (Nevada, Tennessee, South Carolina) combine low friction with low wages — the opposite of what a naive reading of "shortage" would predict.

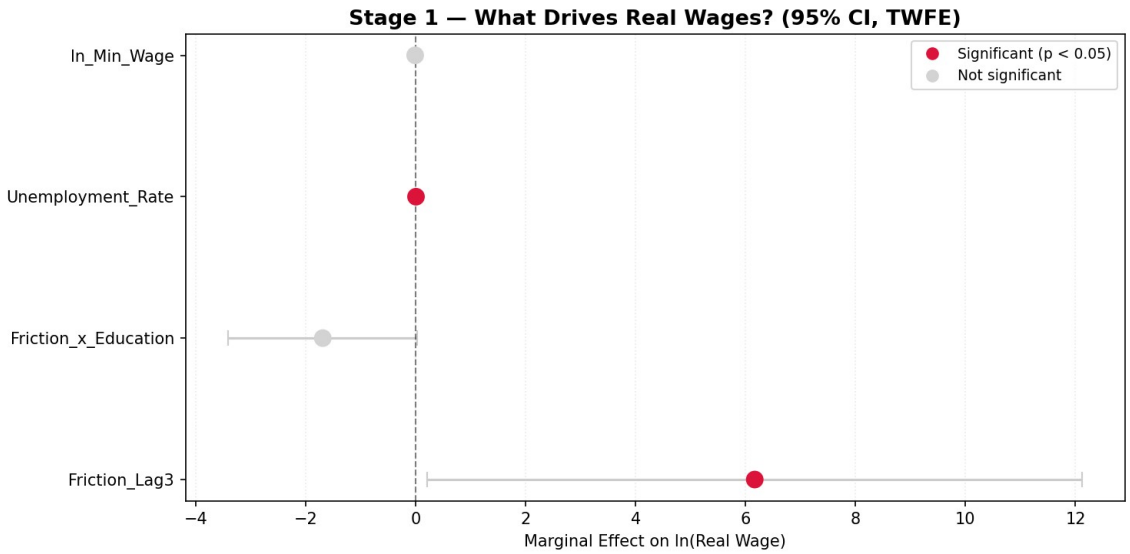


Figure 1 — TWFE marginal effects on log real wages with 95% confidence intervals. The lagged friction proxy is positive and significant; the education interaction is not significant at the 5% level.

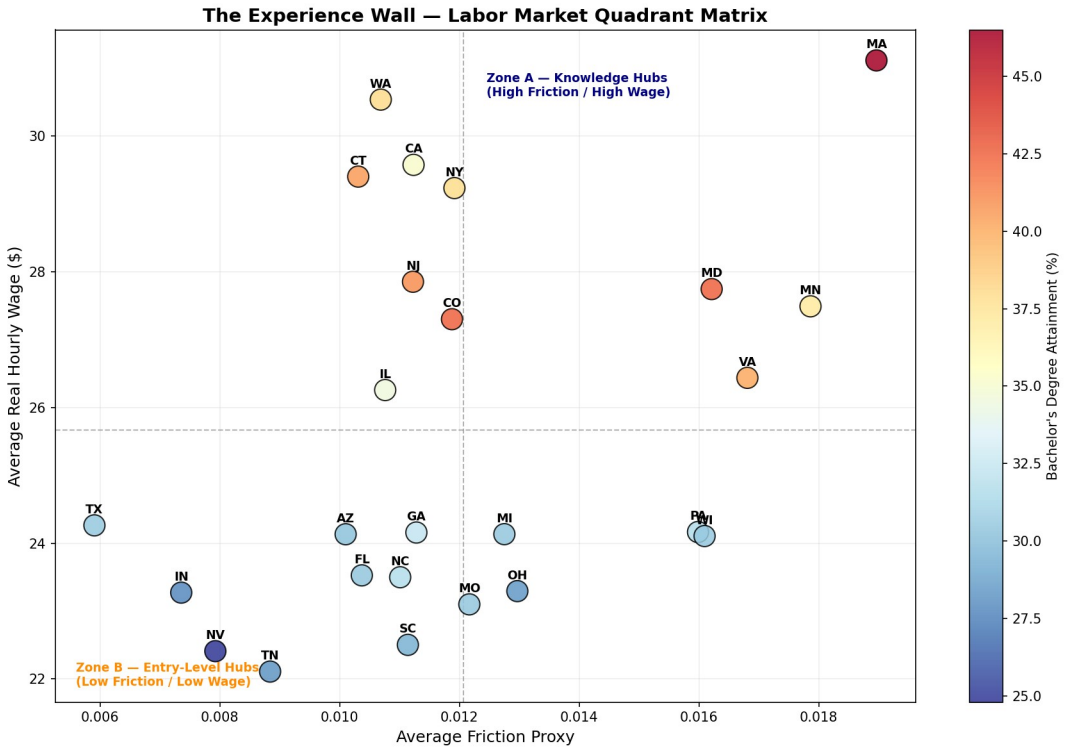


Figure 2 — States by average hiring friction and average real hourly wage, colored by bachelor's attainment. Knowledge hubs sit high-friction/high-wage; entry-level hubs sit low/low.

4.2 Friction does not explain measured tightness (null result)

Table 2 reports Stage 2. Neither the friction proxy (p = 0.61) nor its education interaction (p = 0.43) significantly explains the log tightness ratio. Instead, the leisure-and-hospitality employment share loads enormously and significantly: states with service-heavy economies mechanically show higher openings-per-unemployed-worker. The within-R^2 is 0.24, driven by sector composition rather than hiring friction. In other words, the standard "shortage" indicator is picking up industry mix more than unfilled demand — a caution against reading V/U as a pure demand signal.

Table 2 — Stage 2 (TWFE): log labor-market tightness (openings per unemployed worker). Standard errors clustered by state.

Variable	Coefficient	p-value
Friction_Lag3	-7.8792	0.609
Friction × Education	3.4724	0.426
ln(Min. wage)	-0.3065*	0.072
Leisure & hosp. share	3062.7***	<0.001
Within-R^2	0.2432	

N	2,625	

4.3 No detectable pandemic break (null result)

Table 3 splits the sample at January 2021. In neither subsample is the friction proxy nor its education interaction statistically distinguishable from zero (all p-values above 0.75). Despite the dramatic post-2020 run-up in openings, we cannot detect a structural change in how hiring friction maps to wages. The honest reading is that the test is underpowered — the post-2021 window contains only 900 observations — rather than that nothing changed.

Table 3 — Stage 3: TWFE wage model split at January 2021. All key coefficients insignificant ($p > 0.75$). Standard errors clustered by state.

Variable	**Pre-2021**	**Post-2021**
Friction_Lag3	-0.4240	0.4521
Friction × Education	0.1407	-0.0737
R^2	0.0516	0.1096
N	1,725	900

4.4 Illustrative magnitude: the "inefficiency tax"

As a rough magnitude exercise, we translate the split-sample friction coefficients into an implied annual wage premium per 1,000 employees (friction × β × real wage × 2,080 hours × 1,000 workers). Because the underlying β estimates are statistically insignificant, the resulting dollar figures — on the order of $130,000–$265,000 per 1,000 employees post-2021 in high-friction states — should be read as an illustration of economic scale, not as an estimated causal cost. We include it to show the stakes of the measurement question, not to claim precision the data do not support.

5. Discussion and limitations

Four limitations deserve emphasis. First, the panel covers 25 states, not 50; the sample is the largest states, so results need not generalize to smaller labor markets. Second, educational attainment is a static 2019 cross-section, so the interaction term cannot capture human-capital change over the panel. Third, the friction proxy conflates two distinct phenomena — genuine skill mismatch and "ghost" postings with no hiring intent — which the national ghost-share series suggests became more important after 2021. Fourth, with 25 clusters, the cluster-

robust standard errors rest on fewer groups than the usual rule of thumb recommends; inference should be read accordingly.

The null results are, in our view, as informative as the positive one. That hiring friction does not explain measured tightness — while service-sector concentration does — suggests the conventional shortage narrative leans on a noisy indicator. And that we cannot detect a pandemic break in the friction–wage link, despite the extraordinary post-2020 labor market, is a reminder that striking time-series plots do not always survive a split-sample test.

6. Conclusion

Unfilled job openings predict real wages within states over time, but the relationship reverses in the cross-section — the tightness paradox. At the same time, unfilled openings do not explain the standard tightness ratio, which appears to reflect industry composition more than hiring difficulty. Together, these results argue for more careful measurement of labor demand: counting postings is not the same as observing hiring intent, and policy or business decisions built on headline tightness may be built on a mixture of genuine mismatch, sectoral churn, and postings that were never meant to be filled. Disentangling those components — ideally with posting-level data that distinguishes ghost openings from real ones — is the natural next step.

Data sources

U.S. Bureau of Labor Statistics: Job Openings and Labor Turnover Survey (JOLTS), Local Area Unemployment Statistics (LAUS), Current Employment Statistics (CES), via the BLS public API v2. Federal Reserve Bank of St. Louis (FRED): Consumer Price Index for All Urban Consumers (CPIAUCSL). U.S. Census Bureau: 2019 American Community Survey, educational attainment. U.S. Department of Labor: effective state minimum wage series.